TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

OCTOBER EXAMINATION

BACS2163 SOFTWARE ENGINEERING

MONDAY, 6 OCTOBER 2025 TIME: 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS DEVELOPMENT

BACHELOR IN DATA SCIENCE (HONOURS)

BACHELOR OF SOFTWARE ENGINEERING (HONOURS)

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

This question paper consists of 4 questions on 3 printed pages.

Question 1

A software company is building a desktop accounting system for a small business. The company used the waterfall model to develop the system, but when the client requested major changes midway, the team faced difficulties adapting due to the model's rigid structure.

a) Based on the above situation, identify **TWO (2)** disadvantages of using the waterfall model in this context. (4 marks)

b) Propose a more suitable software process model and explain **TWO (2)** reasons to support your recommendation. (1 + 4 marks)

c) Based on this accounting system project, identify **THREE (3)** non-functional requirements under each of the following categories:

(i) Product Requirements (3 marks)

(ii) Organisational Requirements (3 marks)

(iii) External Requirements (3 marks)

d) Imagine that the system fails to round off currency values correctly due to an error in the calculation logic. Propose any **ONE (1)** type of maintenance required and provide **THREE (3)** reasons to justify your answer. (1 + 6 marks)

[Total: 25 marks]

Question 2

A mobile application is being developed to support a digital canteen system in a smart campus environment. The app allows students to pre-order meals, pay using a virtual wallet, and receive order status updates. However, the digital canteen system experiences heavy user load during peak lunch hours. It is required to quickly handle meal orders, validate payments from virtual wallets, and issue confirmation messages within seconds. Any slowdown in the process may result in duplicate orders, unsuccessful transactions, or extended queues at the collection point.

a) Identify **TWO (2)** advantages and **TWO (2)** disadvantages of using a viewpoint-oriented analysis method in requirements gathering and analysis. Justify each point with a brief explanation. (6 marks)

b) Based on the digital canteen system case study, identify and construct any **SIX (6)** relevant viewpoints that should be considered during the system requirement analysis. (12 marks)

c) Recommend **ONE (1)** suitable type of real-time system that should be implemented to manage these peak time operations effectively. Justify your recommendation with **THREE (3)** supporting reasons based on the system's performance and reliability needs. (1 + 6 marks)

[Total: 25 marks]

Question 3

A logistics software platform is developed in modules: driver tracking, warehouse inventory, order scheduling, and customer notifications. The development team is now performing system integration before delivery.

a) Recommend **ONE (1)** suitable integration testing technique for this modular system. Explain **THREE (3)** advantages of using this approach in the given context. (1 + 6 marks)

b) As a system analyst, you are required to propose **ONE (1)** most suitable system architecture for a modular logistics software platform that integrates the above-mentioned modules in the case study. Support your proposal with **THREE (3)** well-justified reasons for your chosen architecture. (1 + 6 marks)

c) Illustrate the proposed architecture with a clearly labelled diagram showing the interactions among the modules. (5 marks)

d) Discuss **TWO (2)** ways the proposed system architecture contributes to future scalability and maintainability of the logistics platform. (6 marks)

[Total: 25 marks]

Question 4

A university wants to launch a web portal for students to check schedules, submit assignments, and view results. The dashboard must be intuitive, accessible, and responsive.

a) Explain **THREE (3)** UI design principles that must be applied to ensure an effective user experience for the portal. Provide specific examples for each principle based on the above case. (9 marks)

b) You are appointed as the project manager for developing the university web portal. The university has allocated six months and a limited budget for the development. Using Maslow's Hierarchy of Needs, explain **FOUR (4)** ways you can motivate each member in your team to ensure the success of the project. (16 marks)

[Total: 25 marks]